Land 6: Myofiber Forest

Joints and muscle · OpenStax Ch. 9 to 11

Key ideas

Joint Classification

- Structural classes: fibrous (dense connective tissue), cartilaginous (hyaline or fibrocartilage), and synovial (joint cavity).
- Functional classes: synarthrosis (immobile), amphiarthrosis (slightly movable), diarthrosis (freely movable).
- Fibrous joints: sutures of the skull, syndesmoses (radius and ulna interosseous membrane), gomphoses (teeth in sockets).
- Cartilaginous joints: synchondroses use hyaline cartilage (epiphyseal plate, first rib and sternum); symphyses use fibrocartilage (pubic symphysis, intervertebral discs).
- All synovial joints are diarthroses.

Synovial Joints

- Features: articular capsule with outer fibrous layer and inner synovial membrane, joint cavity, synovial fluid, and hyaline articular cartilage.
- Accessory structures: ligaments join bone to bone, tendons join muscle to bone, bursae cushion friction points, menisci and articular discs add fit.
- Six types: plane (intercarpal), hinge (elbow, knee, interphalangeal), pivot (atlantoaxial, proximal radioulnar), condyloid (radiocarpal, metacarpophalangeal), saddle (first carpometacarpal), ball and socket (shoulder, hip).
- Ball and socket joints are multiaxial and allow the greatest range of motion; hinge and pivot are uniaxial.

Body Movements

- Flexion decreases a joint angle; extension increases it. Abduction moves away from the midline; adduction moves toward it.
- Circumduction traces a cone; rotation turns a bone around its own long axis.
- Supination turns the palm anterior; pronation crosses the radius over the ulna so the palm faces posterior.
- Dorsiflexion lifts the foot toward the shin; plantar flexion points the toes. Inversion turns the sole medially; eversion turns it laterally.
- Protraction moves forward and retraction moves back; elevation lifts and depression lowers; opposition brings the thumb to touch the fingertips.

Joint Disorders

- Osteoarthritis is the most common arthritis: degeneration of articular cartilage with age and use, often with bone spurs.
- Rheumatoid arthritis is autoimmune inflammation of the synovial membrane, often symmetric in hands and feet.
- Gouty arthritis results from uric acid crystals deposited in joints, classically the big toe.
- Bursitis is inflammation of a bursa; a sprain is a stretched or torn ligament.

Skeletal Muscle Structure

- Epimysium wraps the whole muscle, perimysium wraps each fascicle, endomysium wraps each fiber.
- Each fiber is a long multinucleated cell with sarcolemma, sarcoplasm, sarcoplasmic reticulum, T tubules, and many myofibrils.
- The sarcomere runs from Z disc to Z disc. The A band is the length of the thick (myosin) filaments; the I band holds only thin (actin) filaments.
- The H zone holds only thick filaments and is centered on the M line.
- Thin filaments carry tropomyosin and troponin, which block myosin binding sites at rest.

Contraction Events

- At the neuromuscular junction the motor neuron releases acetylcholine, which binds receptors on the motor end plate and triggers an action potential.
- Acetylcholinesterase breaks down acetylcholine to end the signal.
- The action potential travels down T tubules, causing the sarcoplasmic reticulum to release calcium, which binds troponin and moves tropomyosin.
- Cross bridge cycle: myosin binds actin, power stroke pulls thin filaments toward the M line, new ATP binding detaches the head, ATP hydrolysis recocks it.
- In contraction the I band and H zone shrink and the A band stays the same; relaxation needs calcium pumped back into the sarcoplasmic reticulum.
- Without ATP myosin cannot detach, causing rigor mortis after death.

Tension, Motor Units, and Energy

- A motor unit is one motor neuron and all the fibers it innervates; small units give fine control, large units give power.
- Recruitment adds more and larger motor units to increase force; muscle tone comes from a few units active at rest.
- A twitch has latent, contraction, and relaxation periods; rapid stimuli cause wave summation and then tetanus.
- Length tension: maximal force at intermediate sarcomere length with optimal overlap.
- ATP sources: creatine phosphate (about the first 15 seconds), anaerobic glycolysis (2 ATP per glucose, makes lactic acid), aerobic respiration in mitochondria (about 36 ATP per glucose).
- Fatigue involves falling ATP and glycogen and rising hydrogen ions that lower pH.

Fiber Types, Cardiac, and Smooth Muscle

- Slow oxidative fibers are red, rich in myoglobin and mitochondria, and fatigue resistant (posture, endurance).
- Fast glycolytic fibers are large, pale, powerful, and fatigue quickly; fast oxidative fibers are intermediate.
- Training causes hypertrophy by adding myofibrils; disuse or denervation causes atrophy.
- Cardiac muscle is striated and involuntary, with branched cells, one or two nuclei, intercalated discs, and a long refractory period that prevents tetanus.
- Smooth muscle is nonstriated and involuntary, uses calmodulin instead of troponin, anchors filaments to dense bodies, and can stretch widely.

Muscle Organization and Major Muscles

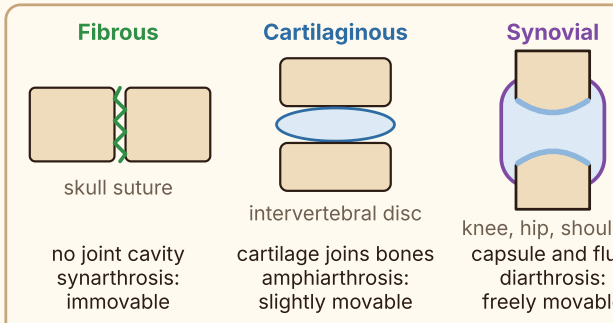
- Prime mover (agonist) does the movement, synergists assist, antagonists oppose, fixators stabilize the origin. The origin stays fixed and the insertion moves.
- Fascicle arrangements: parallel, convergent, pennate (uni, bi, multi), and circular (sphincters). Pennate muscles pack in more fibers for more force.
- Levers: first class has the fulcrum in the middle (head tilting back on the atlas); third class, most common, has effort between fulcrum and load (biceps at the elbow).
- Names describe location, shape, size, fiber direction, number of origins, attachments, or action (deltoid, rectus, triceps, sternocleidomastoid).
- Head and neck: orbicularis oculi closes the eye, masseter and temporalis elevate the mandible, sternocleidomastoid flexes and rotates the neck. Diaphragm is the main muscle of inspiration.
- Limbs: deltoid abducts the arm, biceps brachii flexes and triceps brachii extends the elbow, quadriceps extend the knee, hamstrings flex the knee, gastrocnemius and soleus plantar flex, tibialis anterior dorsiflexes.

Villager wisdom

- Joints are named two ways: by what holds them together (fibrous, cartilaginous, synovial) and by how much they move (synarthrosis, amphiarthrosis, diarthrosis).
- In the sliding filament dance, no filament gets shorter. Myosin heads grab actin and pull it toward the middle, so the whole sarcomere shortens.
- Calcium is the key that unlocks actin. In skeletal and cardiac muscle it binds troponin; in smooth muscle it binds calmodulin.
- For a quick burst, creatine phosphate refills ATP first, then anaerobic glycolysis helps, and aerobic respiration carries you through the long journey.
- The origin stays put and the insertion moves. The agonist does the work while the antagonist on the opposite side relaxes and lets it happen.

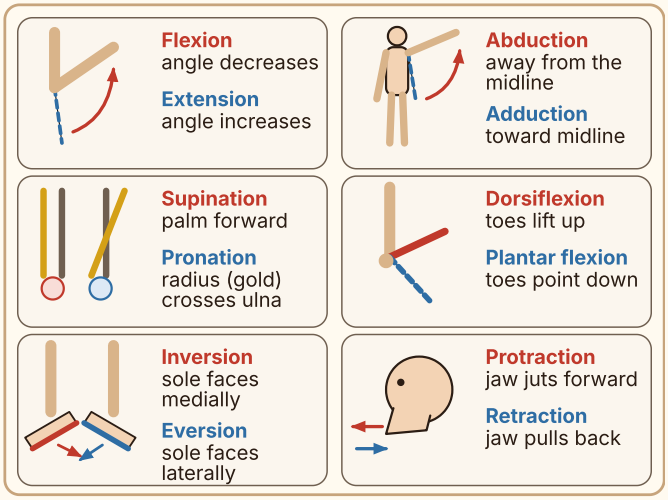
Diagrams

Joint classification



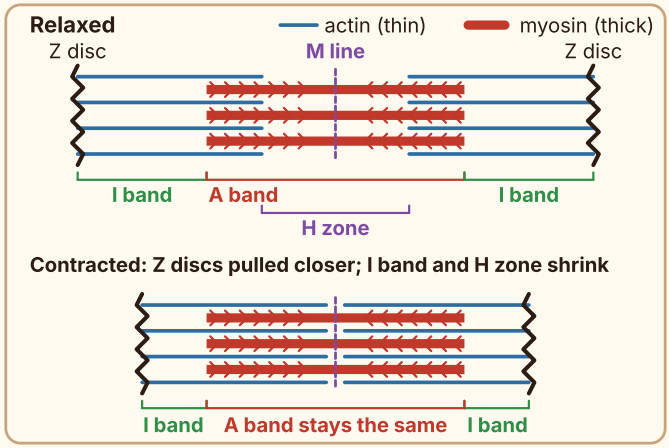
Structurally, joints are fibrous, cartilaginous or synovial; functionally they range from immovable (synarthrosis) to freely movable (diarthrosis), which all synovial joints are.

Movements at synovial joints



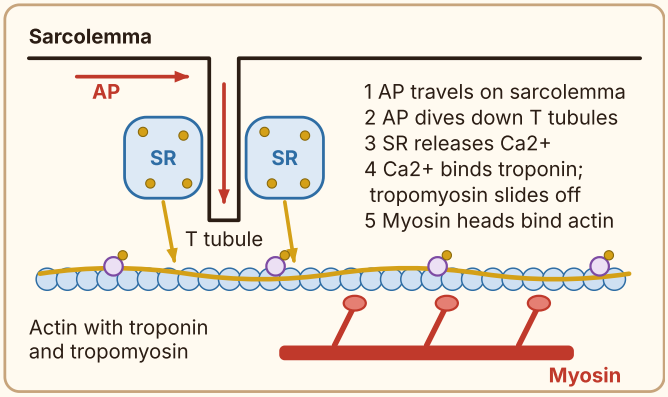
Movements come in opposing pairs: flexion closes a joint angle and extension opens it, abduction moves a limb away from the midline, and pronation turns the palm back as the radius crosses the ulna.

The sarcomere



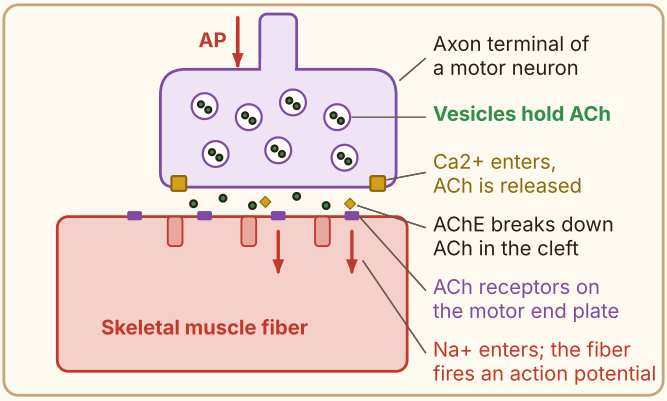
During contraction thin filaments slide toward the M line, pulling Z discs closer; the I band and H zone shorten while the A band (thick filament length) stays the same.

Excitation contraction coupling



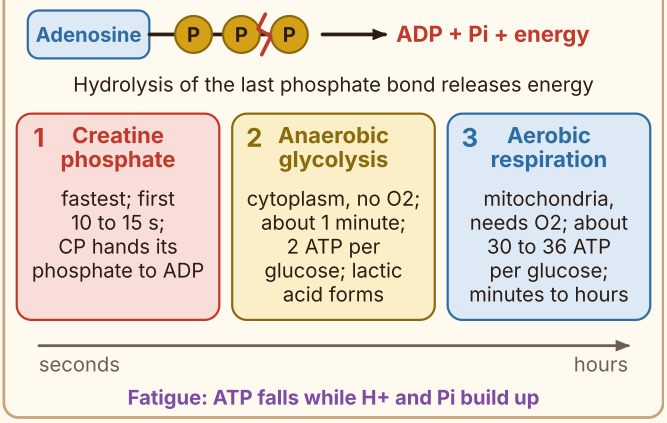
The muscle action potential runs down T tubules, the SR releases Ca2+, and Ca2+ binds troponin so tropomyosin uncovers actin binding sites for myosin heads.

Neuromuscular junction



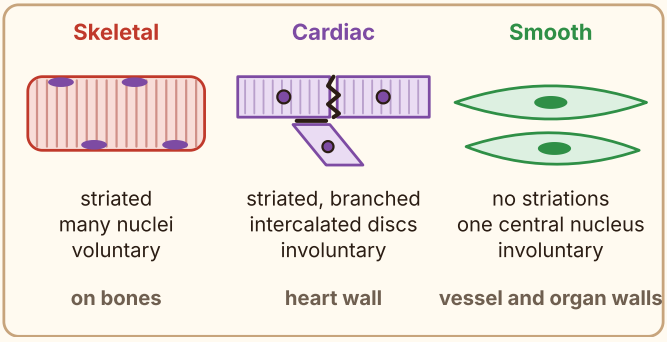
An action potential opens Ca2+ channels in the axon terminal, ACh is released, binds receptors on the motor end plate and opens Na+ channels; AChE then breaks ACh down.

ATP and muscle energy systems



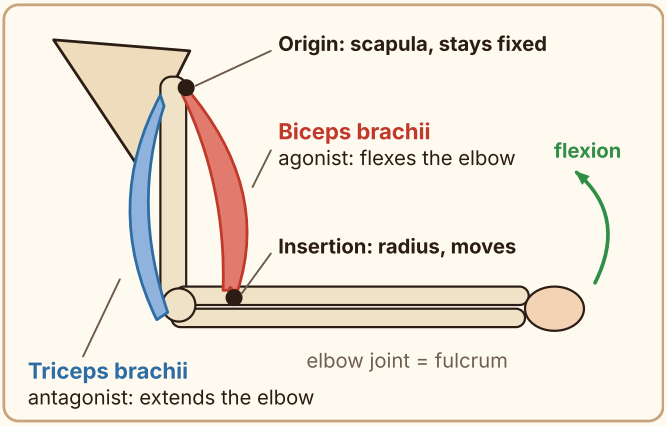
Muscle first rebuilds ATP from creatine phosphate, then from anaerobic glycolysis for about a minute, and for longer activity from aerobic respiration in the mitochondria, which yields far more ATP.

Three types of muscle tissue



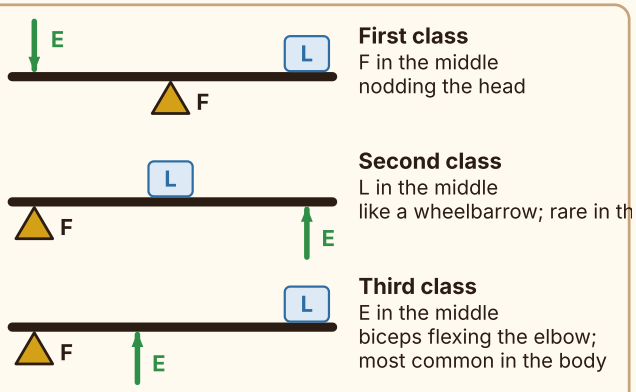
Skeletal muscle is striated, multinucleate and voluntary; cardiac muscle is striated, branched and joined by intercalated discs; smooth muscle is spindle shaped with no striations.

Agonist, antagonist, origin, insertion



The prime mover (agonist) produces the movement while the antagonist opposes it; the origin stays fixed and the insertion is pulled toward it.

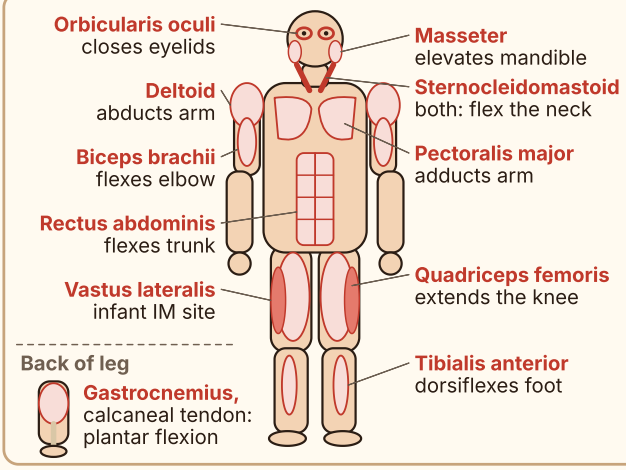
Lever systems



E = effort, F = fulcrum (joint), L = load

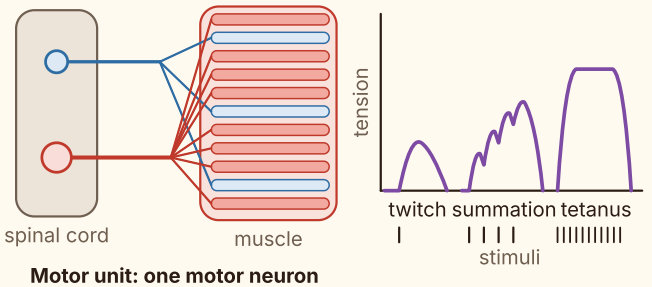
Levers are classed by what sits in the middle: fulcrum (first), load (second) or effort (third); most body levers, like the biceps at the elbow, are third class.

Major skeletal muscles



Each muscle is named for a feature such as shape (deltoid), location or action; the quadriceps extends the knee and the gastrocnemius pulls on the calcaneal tendon to plantar flex the foot.

Motor units and muscle tension



Motor unit: one motor neuron
plus all the fibers it supplies

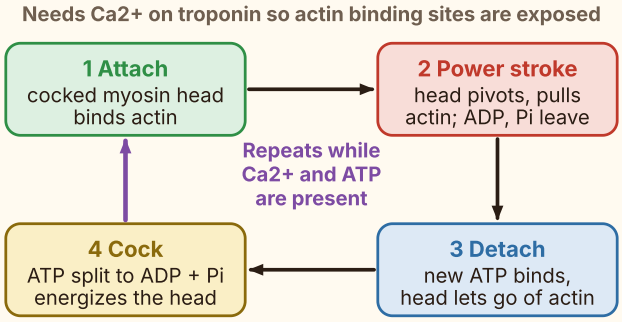
Small unit (blue): fine control
Large unit (red): more force

More force: recruit
more and larger units

Tone: a few units active even at rest

A motor unit is one motor neuron and every fiber it supplies; the nervous system adds force by recruiting more and larger units and by stimulating faster, from twitch to summation to tetanus.

The cross bridge cycle



No ATP (rigor mortis): myosin cannot detach, so heads stay locked to actin and muscle is rigid.

Each cycle pulls actin a short step toward the M line; ATP is needed to detach the myosin head, which is why muscles lock in rigor mortis when ATP runs out.

Self check

1. **A nurse gives an intramuscular injection into the lateral thigh of an infant. Which muscle receives it?**
    A. Vastus lateralis
    B. Gluteus maximus
    C. Rectus abdominis
    D. Gastrocnemius
2. **The biceps brachii attaches to the scapula and the radius. During elbow flexion the scapula stays still. What is the radial attachment called?**
    A. Insertion
    B. Origin
    C. Fulcrum
    D. Belly
3. **When both sternocleidomastoid muscles contract together, what movement results?**
    A. Flexion of the neck
    B. Extension of the neck
    C. Elevation of the scapulae
    D. Depression of the mandible
4. **A tooth held in its bony socket by the periodontal ligament is an example of which joint?**
    A. Syndesmosis
    B. Symphysis
    C. Gomphosis
    D. Synchondrosis
5. **Why can cardiac muscle not go into tetanus the way skeletal muscle can?**
    A. It has a long refractory period
    B. It has no calcium stores
    C. It lacks sarcomeres
    D. It is under voluntary control
6. **According to the sliding filament model, what happens during contraction?**
    A. Thin filaments slide toward the M line
    B. Thick filaments shorten
    C. Thin filaments shorten
    D. The A band narrows
7. **Tilting the head back at the joint between the skull and atlas uses a lever with the fulcrum between the effort and the load. Which class is it?**
    A. Second class
    B. Third class
    C. First class
    D. Not a lever system

8. **When the biceps brachii flexes the elbow, the elbow joint is the fulcrum and the biceps pulls on the radius between the elbow and a weight held in the hand. Which lever class is this?**
    A. First class
    B. Third class
    C. Second class
    D. Pulley system
9. **The atlantoaxial joint lets you shake your head to say no. What type of synovial joint is it?**
    A. Hinge
    B. Condyloid
    C. Plane
    D. Pivot
10. **Jutting the lower jaw forward to push the chin out is called:**
    A. Retraction
    B. Elevation
    C. Protraction
    D. Depression
11. **A soccer player twists the knee and now has pain with clicking. Imaging shows a torn C shaped fibrocartilage pad. What is injured?**
    A. A bursa
    B. The fibrous capsule
    C. The synovial membrane
    D. A meniscus
12. **During repeated maximal lifts, a weightlifter's force output steadily declines. Which change contributes most to this fatigue?**
    A. Falling ATP and rising hydrogen ions
    B. An increase in stored glycogen
    C. Extra oxygen bound to myoglobin
    D. A buildup of creatine phosphate

Answer key

1. **A.** Vastus lateralis. The vastus lateralis is a large quadriceps muscle on the lateral thigh and is the preferred injection site in infants. The gluteus maximus is in the buttock, not the lateral thigh.
2. **A.** Insertion. The insertion is the attachment on the bone that moves, here the radial tuberosity. The origin is the more stationary attachment, which for the biceps is on the scapula.
3. **A.** Flexion of the neck. Acting together, the two sternocleidomastoids flex the neck, bringing the chin toward the chest; acting alone, one rotates the head to the opposite side. Elevation of the scapulae is done by the trapezius and levator scapulae.
4. **C.** Gomphosis. A gomphosis is a fibrous joint that anchors a tooth in its socket, and it is functionally a synarthrosis. A syndesmosis is also fibrous, but it joins bones with a ligament or interosseous membrane, as between the radius and ulna.
5. **A.** It has a long refractory period. The long refractory period of cardiac muscle lasts nearly as long as the contraction, so a new contraction cannot begin until relaxation is underway. Cardiac muscle is striated and has sarcomeres, so lacking them is not the reason.

6. **A.** Thin filaments slide toward the M line. Myosin heads pull actin filaments toward the center of the sarcomere, so the sarcomere shortens while every filament keeps its length. The A band equals the length of the thick filaments and stays the same; the I band and H zone narrow.
7. **C.** First class. In a first class lever the fulcrum lies between the effort (posterior neck muscles) and the load (weight of the face). A third class lever has the effort between the fulcrum and the load, as at the elbow.
8. **B.** Third class. When the effort is applied between the fulcrum and the load, it is a third class lever, the most common type in the body; it favors speed and range of motion over force. In a second class lever the load sits between the fulcrum and the effort, as in a wheelbarrow.
9. **D.** Pivot. In a pivot joint a rounded process rotates within a ring, as the dens of the axis turns within the atlas. A hinge joint allows only flexion and extension, not rotation.
10. **C.** Protraction. Protraction moves a structure anteriorly in a horizontal plane, as with the mandible or scapulae. Retraction is the opposite, pulling the structure back posteriorly.
11. **D.** A meniscus. The medial and lateral menisci are C shaped fibrocartilage pads that deepen the knee joint and are often torn by twisting. A bursa is a fluid filled sac, not a firm fibrocartilage pad.
12. **A.** Falling ATP and rising hydrogen ions. Fatigue develops as ATP falls and hydrogen ions accumulate, lowering pH and impairing contraction. Glycogen and creatine phosphate stores are used up during intense work, not increased.

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